MIL-E-1/125A
19 February 1970
SUPERSEDING
MIL-E-1/125
30 March 1953

MILITARY SPECIFICATION SHEET

ELECTRON TUBES, RECEIVING

TYPES 6SR7 AND 12SW7 1/

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Duplex diode, low-Mu triode

Ⓐ

Outline	—	8-1 (EIA)
Base	—	B8-21
Envelope	—	MT8
Cathode	—	Coated unipotential
Base connections:		

Pin No.	—	1	2	3	4	5	6	7	8
Element	—	sh	g	k	2a	1a	a	h	h
					diodes		triode		

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	E1b	E2b	I1b	I2b	Rg	Pp	Ⓐ Ehk	Ⓐ Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	mAdc	mAdc	Meg	W	v	ft
Type 6SR7:											
Maximum:	6.9	275	—	—	—	1.0	1.0	—	2.75	100	(See
Minimum:	5.7	—	—	—	—	—	—	—	—	—	note 2)
Type 12SW7:											
Maximum:	13.9	275	—	—	—	1.0	1.0	—	2.75	100	(See
Minimum:	11.3	—	—	—	—	—	—	—	—	—	note 2)
TEST CONDITION (1):											
Type 6SR7:	6.3	250	-9	0	0	—	—	—	—	—	—
Type 12SW7:	12.6	250	-9	0	0	—	—	—	—	—	—
TEST CONDITION (2):											
Type 12SW7:	12.6	28	—	—	—	—	—	10(max)	—	—	—

GENERAL:

Qualification - Required

1/ See note 1

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1266	Total grid current	Test condition (1) (see note 3)	1.0	II	Ic	.0	-1.0	μAdc
1306	Transconductance	Test condition (1)	1.0	II	Sm	1,500	2,300	μmhos
1321	AC amplification Type 12SW7	Test condition (2); Rp = 50,000 ohms; Esig = 0.2 Vac	1.0	II	Ep	1.6	—	Vac
1246	Audio frequency noise	Test condition (1); Esig = 70 mVac; Rp = 10,000 ohms	1.0	II	—	—	—	—
1256	Electrode current (anode) (triode)	Test condition (1)	1.0	II	Ib	6.3	12.7	mAdc
1231	Emission (triode)	Eb = Ec = 30 Vdc (see note 3)	1.0	II	Is	30	—	mAdc
1231	Emission (diodes)	E1b = 10 Vdc	1.0	II	I1s	0.8	—	mAdc
		E2b = 10 Vdc	1.0	II	I2s	0.8	—	mAdc
	Quality conformance inspection, part 2							
1301	Heater current							
	Type 6SR7		—	—	If	275	325	mA
	Type 12SW7		—	—	If	138	162	mA
(A) 1336	Heater-cathode leakage		—	—	Ihk	—	20	μAdc
1316	Amplification factor	Test condition (1)	—	—	Mu	14	18	—
(A) 1331	Direct-interelectrode capacitance		—	—	Cgp	1.6	3.0	pF
					Cin	1.8	3.6	pF
					Cout	2.4	3.4	pF
1031	Low-frequency vibration	Test condition (1); Rp = 2,000 ohms	—	—	Ep	—	150	mVac
(A) 1105	Permanence of marking		—	—	—	—	—	—
	Quality conformance inspection, part 3							
1501	Intermittent life	Test condition (1); Group A; I1b = I2b = 1.0 mAdc; Ehk = 100 V	—	—	—	—	—	—
—	Intermittent-life test end points (500 hours)	Transconductance	—	—	Sm	1,230	—	μmhos
		Emission(diodes)	—	—	I1s	0.3	—	mAdc
					I2s	0.3	—	mAdc

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NOTES:

1. Tube types 6SR7GT, 12SR7, 12SR7GT, and 12SW7GT are hereby deleted from this specification sheet. For replacement purposes, procure tube type 6SR7 for tube type 6SR7GT and tube type 12SW7 for tube types 12SR7, 12SR7GT, and 12SW7GT.

2. See "Reduced pressure (altitude) rating", and altitude maximum peak voltage.

3. This test to be performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL
Navy -
Air Force - 11, 70, 85
DSA - ES

User activities:
Army - ME, AV, WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2401-11)

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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